

Amjad S Althabteh

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Summary

C++ Graphics / Engine Developer specializing in real-time rendering, GPU optimization, and engine architecture. Built OpenGL rendering engines with GPU-driven pipelines, custom allocators, and performance profiling tools. Experience developing gameplay systems and tools in Unreal Engine 5.

Education

B.S. in Computer Science — Wayne State University
Detroit, MI | Expected 2026 | GPA: 3.65/4.00

Experience

Graphics Engineer *Immersalab Graphics* *Detroit, MI* **Sep 2025 - Dec 2025**

- Architected a C++20/OpenGL real-time rendering engine with GPU-driven pipelines and frame-time stability optimizations.
- Built internal UI tools and a JSON asset pipeline for flexible scene configuration.
- Implemented data-driven optimizations to balance GPU workloads.
- Developed GPU profiling and frame-timing tools to analyze rendering performance.

Unreal C++ Dev *Indie Game (NOA)* *Ann Arbor, MI* **Apr 2025 - Aug 2025**

- Developed gameplay systems in C++20 and Unreal Blueprints integrating animation and narrative logic.
- Produced a 90-second real-time cinematic using MetaHuman, Control Rig, and Sequencer; demo available in portfolio.
- Built animation control systems using C++ and Animation Blueprints for smooth blending.

Software Development Intern *Rare Munchies* *Ann Arbor, MI* **Sep 2023 - Apr 2024**

- Integrated UI components using **TypeScript**, improving page transitions and interaction flow.
- Identified and resolved **cross-browser compatibility issues**, reducing debugging time and unblocking delivery
- Optimized client-side rendering logic and state management patterns to reduce unexpected UI re-renders.

Undergraduate Researcher — Orbital Collision Modeling
Wayne State University — Detroit, MI | Feb 2025 – Present

- Modeled satellite trajectories using numerical integration and N-body gravitational dynamics.
- Simulated **orbital collision probabilities** using spatial partitioning techniques.

Projects

Glint 3D – C++ 3D Rendering Engine *glint-rendering*

- Optimized a C++17/OpenGL rendering engine with improved GPU workload distribution and lighting systems.
- Implemented frustum culling and instanced rendering to reduce draw calls in large scenes.
- Designed a custom C++20 memory pool allocator to reduce heap allocations and runtime overhead.

3D Space Simulation Engine – C++ / OpenGL *space-simulations*

- Developed a C++17/OpenGL rendering engine supporting dynamic lighting and GPU workload balancing.
- Designed modular engine systems for rendering, physics, math, and entity management.
- Implemented a custom math library (Vec3, Mat4) and camera system with projection/view matrices.

HFT OrderBook Engine — C++20

- Implemented a high-performance price-time FIFO order matching engine supporting bid/ask depth tracking.
- Designed low-latency cache-aligned data structures using modern C++20.

Traceforge – C++ Runtime Stack Trace & Diagnostics Engine

- Developed a C++23 runtime diagnostics system for stack trace parsing and debugging analysis.
- Implemented regression test generation and memory tracking utilities.

Skills

- **Programming:** C++ (C++11–C++23), C, C#, Python, TypeScript, Java, SQL
- **Graphics & Engines & Tools:** OpenGL, Unreal Engine 4–5, Unity, Godot, Git, CMake, VS code, Git